Proposal of a New Software Architecture for Interoperability to Improve the Communication in the Edge Layer of a Smart IoT Ecosystem

Juan Moreno-Motta
Universidad Nacional Mayor de San Marcos

Felipe Moreno-Vera
Universidad Católica San Pablo

Frank Moreno-Vera
Universidad Nacional de Ingeniería

SERP4IoT Workshop · ICSE 2019

Motivation

- **Problem:** IoT devices generate heterogeneous data. Interoperability means more than adding or removing devices: the data must also be standardized and optimized to transmit.
- **Why it is hard:** devices differ in brands, hardware, protocols, message encoding, programming languages and data structures [8, 9].
- **Our proposal:** a new software architecture pattern for interoperability in the edge layer of an IoT ecosystem.

1.7 B
estimated increase in IoT devices [1]

150 EB
of device storage surpassed in 2017 [2]

72
agro and environment IoT projects reviewed [4]

1 of 9
requirements: interoperability, the most difficult [5]

How can a secure system manage interoperability without complications?

Related Work

Binary encoder
AAL-IoTSys: heterogeneous low-power devices on a WSN [10]

JSON-LD
Semantic web descriptions that link objects and properties [11]

REST + JSON
Web of Things API; microservices vs. monolith [12]

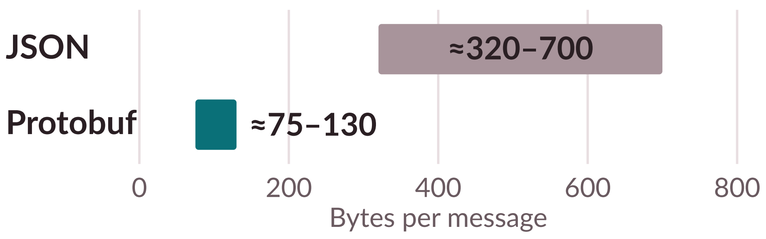
SOAP vs. REST
Tried to improve the scalability of interoperability, without success [13, 14]

Fog servers
Middle servers between the edge layer and cloud services [15]

Routing gateway
Connects devices, fog nodes and cloud servers over 5G [16]

Why Protocol Buffers

Message size: Protobuf vs. JSON



Range of average message length, read approximately from Fig. 3 of the paper (Lysogor et al. [17]).

≈4–5×
smaller messages on LoRa sensors, per the values read from the figure.

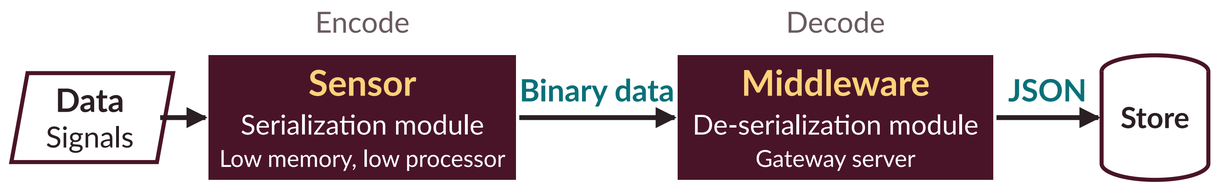
Protobuf's binary format has **better performance and less memory use**, and serializes many more messages per second on any communication protocol [19].

Common IoT messaging protocols [18]

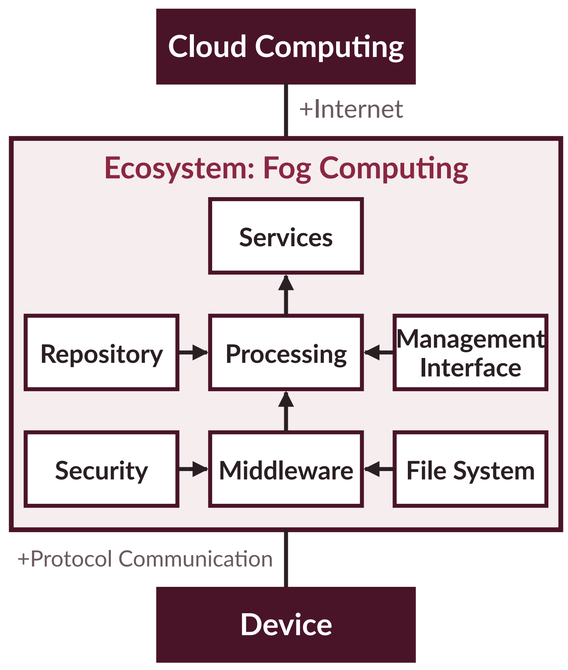
	MQTT	CoAP	AMQP	HTTP
Architecture	Client/Broker	Client/Server or Client/Broker	Client/Broker or Client/Server	Client/Server
Header size	2 bytes	4 bytes	8 bytes	Undefined
Transport	TCP (MQTT-SN can use UDP)	UDP, SCTP	TCP, SCTP	TCP
Security	TLS/SSL	DTLS, IPSec	TLS/SSL, IPSec, SASL	TLS/SSL
Encoding	Binary	Binary	Binary	Text

Proposal: the Ecosystem

An **Ecosystem** is the process that communicates data between devices, IoT solutions or frameworks through **encode-decode serialization**, based on a microservices architecture.



Fog Ecosystem: 7 Components

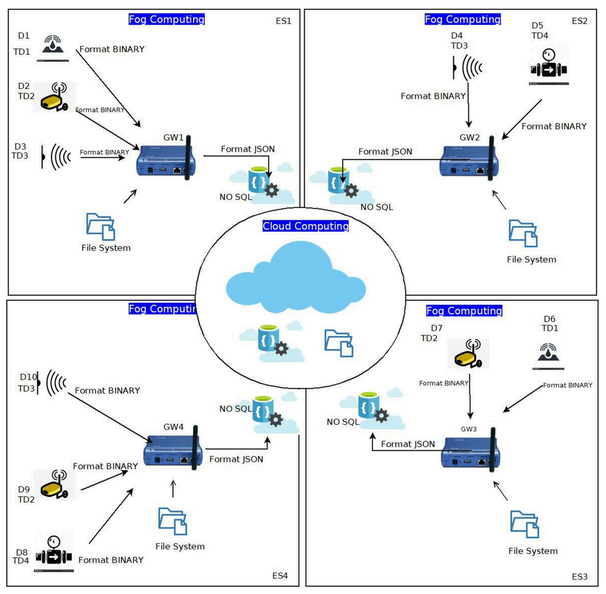


Proposed architecture (redrawn from Fig. 7).

- **Middleware:** receives binary frames; uses Processing and the File System.
- **Processing:** identifies the device type and origin, and processes the binary data.
- **Security:** validates that connecting devices have the required authorization.
- **Repository:** persists de-serialized frames as JSON.
- **File System:** the ecosystem's logger; also stores decoded data.
- **Management Interface:** registers devices, their types, middleware, ecosystems.
- **Services:** let external devices read sensors or send actions to actuators.

Devices encode and decode with Protocol Buffers; **Cloud Computing** is the final repository for analyzing pre-processed data.

Multiple Solutions



Proposed for multiple solutions (Fig. 8).

- **Devices** (D) of each type (TD) send **binary** frames to their gateway (GW).
- The **gateway** stores **JSON** in NoSQL, using the File System.
- Four **ecosystems** (ES) share one cloud.

Conclusions & Future Work

- A new interoperability architecture with **Fog and Edge computing**, integrable with other ecosystems, aiming at scalability and at identifying data origin.
- **Protocol Buffers** and a standardized format seek better performance and less latency.
- **In progress:** a real-time proof of concept (temperature, humidity, sound and monoxide sensors, Raspberry Pi 3, Arduino, SQLite, Firebase) to measure latency, compatibility and processing speed.

Acknowledgment: supported by grant 234-2015-FONDECYT (Master Program) from CienciActiva of CONCYTEC-PERU, and by the Scotiabank Master Program Grant (Juan Moreno).

1. O. Kleine. CoAP Endpoint Identification. IEEE iThings, 2014.
2. U. Jaimini. Multimodal IoT and EMR based Smart Health Application for Asthma Management in Children. 2017.
3. D. Guinard, V. Trifa. Building the Web of Things. Manning, 2016.
4. J. M. Talavera et al. Review of IoT applications in agro-industrial and environmental fields. 2017.
5. P. Woznowski et al. SPHERE: A Sensor Platform for Healthcare in a Residential Environment.
6. A. Elsts et al. Internet of Things for Smart Homes: Lessons Learned from the SPHERE Case Study. 2017.
7. A. Alkhalil, R. Ramadan. IoT Data Provenance Implementation Challenges. 2017.
8. P. Pace et al. IoT platforms interoperability for Active and Assisted Living Healthcare services support. 2017.
9. N. Madaan et al. Data integration in IoT ecosystem: Information linkage as a privacy threat. 2017.
10. D. Yacchirema et al. Enable IoT Interoperability in Ambient Assisted Living. 2017.
11. D. Androcec et al. Interoperability and Lightweight Security for Simple IoT Devices. 2017.
12. L. Sun et al. An Open IoT Framework Based on Microservices Architecture. 2017.
13. N. Lim et al. Providing Interoperability for Resource Access Using Web Services. 2010.
14. S. Malik, D.-H. Kim. A Comparison of RESTful vs. SOAP Web Services in Actuator Networks. 2017.
15. S. W. Kum et al. Design of Fog Computing based IoT Application Architecture. 2017.
16. T. H. Luan et al. Fog Computing: Focusing on Mobile Users at the Edge. 2016.
17. I. Lysogor et al. Survey of Data Exchange Formats for Heterogeneous LPWAN-Satellite IoT Networks. 2018.
18. N. Naik. Choice of effective messaging protocols for IoT systems: MQTT, CoAP, AMQP and HTTP. IEEE ISSE, 2017.
19. B. Petersen et al. Smart Grid Serialization Comparison. 2017.